

# Incident Report

|                     |                                                                   |                 |                                                   |
|---------------------|-------------------------------------------------------------------|-----------------|---------------------------------------------------|
| Device ID:          | DEV-001                                                           | Sector:         | 1                                                 |
| Device Name:        | AEGIS-PUMP-01                                                     | Timestamp:      | 2026-05-09 09:44:15                               |
| Incident Signature: | 8c1b2db7b8fa9ec1a2c30fdb0a1daaf5b7e4943ce1c734cc7f4702a8942d572b1 | Severity:       | Critical                                          |
| Trust Score:        | 14.9%                                                             | Attack Pattern: | Data Exfiltration (large payloads + high entropy) |

## Executive Summary

A critical anomaly was detected on the digital twin for the device. The system automatically generated this report to capture a snapshot of the observed behavior, analysis results, and recommended next steps for investigation and remediation.

## Digital Twin Analysis

The LSTM autoencoder measured the reconstruction error and compared the incoming traffic features to the expected baseline. These discrepancies were used to compute a Trust Score and to highlight the most anomalous data attributes.

| Metric                     | Value   |
|----------------------------|---------|
| Trust Score                | 14.9%   |
| Reconstruction Error (MSE) | 0.47143 |
| Jensen-Shannon Divergence  | 0.40801 |

## Top Anomaly Contributing Features

| Feature     | Delta |
|-------------|-------|
| Entropy     | 0.700 |
| Packet Size | 0.600 |
| Symmetry    | 0.600 |

## Behavioral Timeline

- 15:14:14: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 15:14:05: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 15:13:59: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 15:13:54: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 15:13:50: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 15:13:46: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 15:13:41: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471
- 15:13:37: Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

**15:13:32:** Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

**15:13:28:** Anomalous flow detected! Trust dropped to 14.9%. MSE: 0.471

Possible Attack Pattern

Data Exfiltration (large payloads + high entropy)

Risk Assessment

Based on the observed anomaly magnitude and the estimated severity, this incident is considered **CRITICAL**. The device should be treated as compromised until a full investigation confirms otherwise.

Recommended Remediation Actions

- Isolate the device from the network (quarantine).
- Collect full packet captures and system logs for post-incident analysis.
- Perform a firmware integrity check and verify configuration drift.
- Rotate credentials and revoke all active sessions for the device.
- Update IDS/IPS rules to detect similar behavior in the future.

Digital Evidence Snapshot

| Baseline Features          | Current Features           |
|----------------------------|----------------------------|
| 0.400, 0.500, 0.300, 0.600 | 1.000, 0.000, 1.000, 0.000 |

Incident Signature

8c1b2db7b8fa9ec1a2c30fdb0a1daafd4704943ce1c734c87740038942d572b1